JavaScript is disabled on your browser.

Java SE 27 & JDK 27

Class ZipEntry

java.lang.Object
    java.util.zip.ZipEntry

All Implemented Interfaces:

Cloneable

Direct Known Subclasses:

JarEntry

```
public class ZipEntry
extends Object
implements Cloneable
```

This class is used to represent a ZIP file entry.

Since:

1.1

Field Summary

Fields

Modifier and Type	Field	Description
static final int	CENATT	Central directory (CEN) header internal file attributes field offset.
static final int	CENATX	Central directory (CEN) header external file attributes field offset.
static final int	CENCOM	Central directory (CEN) header comment length field offset.
static final int	CENCRC	Central directory (CEN) header uncompressed file crc-32 value field offset.
static final int	CENDSK	Central directory (CEN) header disk number start field offset.
static final int	CENEXT	

		Central directory (CEN) header extra field length field offset.
static final int	CENFLG	Central directory (CEN) header encrypt, decrypt flags field offset.
static final int	CENHDR	Central directory (CEN) header size in bytes (including signature).
static final int	CENHOW	Central directory (CEN) header compression method field offset.
static final int	CENLEN	Central directory (CEN) header uncompressed size field offset.
static final int	CENNAM	Central directory (CEN) header filename length field offset.
static final int	CENOFF	Central directory (CEN) header LOC header offset field offset.
static final long	CENSIG	Central directory (CEN) header signature.
static final int	CENSIZ	Central directory (CEN) header compressed size field offset.
static final int	CENTIM	Central directory (CEN) header modification time field offset.
static final int	CENVEM	Central directory (CEN) header version made by field offset.
static final int	CENVER	Central directory (CEN) header version needed to extract field offset.

static final int	**DEFLATED**	Compression method for compressed (deflated) entries.
static final int	**ENDCOM**	End of central directory (END) header ZIP file comment length field offset.
static final int	**ENDHDR**	End of central directory (END) header size in bytes (including signature).
static final int	**ENDOFF**	End of central directory (END) header offset for the first CEN header field offset.
static final long	**ENDSIG**	End of central directory (END) header signature.
static final int	**ENDSIZ**	End of central directory (END) header central directory size in bytes field offset.
static final int	**ENDSUB**	End of central directory (END) header number of entries on this disk field offset.
static final int	**ENDTOT**	End of central directory (END) header total number of entries field offset.
static final int	**EXTCRC**	Extra local (EXT) header uncompressed file crc-32 value field offset.
static final int	**EXTHDR**	Extra local (EXT) header size in bytes (including signature).
static final int	**EXTLEN**	

		Extra local (EXT) header uncompressed size field offset.
static final long	**EXTSIG**	Extra local (EXT) header signature.
static final int	**EXTSIZ**	Extra local (EXT) header compressed size field offset.
static final int	**LOCCRC**	Local file (LOC) header uncompressed file crc-32 value field offset.
static final int	**LOCEXT**	Local file (LOC) header extra field length field offset.
static final int	**LOCFLG**	Local file (LOC) header general purpose bit flag field offset.
static final int	**LOCHDR**	Local file (LOC) header size in bytes (including signature).
static final int	**LOCHOW**	Local file (LOC) header compression method field offset.
static final int	**LOCLEN**	Local file (LOC) header uncompressed size field offset.
static final int	**LOCNAM**	Local file (LOC) header filename length field offset.
static final long	**LOCSIG**	Local file (LOC) header signature.
static final int	**LOCSIZ**	Local file (LOC) header compressed size field offset.
static final int	**LOCTIM**	

		Local file (LOC) header modification time field offset.
static final int	LOCVER	Local file (LOC) header version needed to extract field offset.
static final int	STORED	Compression method for uncompressed entries.

Constructor Summary

Constructors

Constructor	Description
ZipEntry(String name)	Creates a new ZIP entry with the specified name.
ZipEntry(ZipEntry e)	Creates a new ZIP entry with fields taken from the specified ZIP entry.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
Object	clone()	Returns a copy of this entry.
String	getComment()	Returns the comment string for the entry.
long	getCompressedSize()	Returns the size of the compressed entry data.
long	getCrc()	Returns the CRC-32 checksum of the uncompressed entry data.
FileTime	getCreationTime()	Returns the creation time of the entry.

byte[]	**getExtra**()	Returns the extra field data for the entry.
FileTime	**getLastAccessTime**()	Returns the last access time of the entry.
FileTime	**getLastModifiedTime**()	Returns the last modification time of the entry.
int	**getMethod**()	Returns the compression method of the entry.
String	**getName**()	Returns the name of the entry.
long	**getSize**()	Returns the uncompressed size of the entry data.
long	**getTime**()	Returns the last modification time of the entry.
LocalDateTime	**getTimeLocal**()	Returns the last modification time of the entry in local date-time.
int	**hashCode**()	Returns the hash code value for this entry.
boolean	**isDirectory**()	Returns true if this is a directory entry.
void	**setComment**(String comment)	Sets the optional comment string for the entry.
void	**setCompressedSize**(long csize)	Sets the size of the compressed entry data.
void	**setCrc**(long crc)	Sets the CRC-32 checksum of the uncompressed entry data.
ZipEntry		

	setCreationTime(FileTime time)	Sets the creation time of the entry.
void	**setExtra**(byte[] extra)	Sets the optional extra field data for the entry.
ZipEntry	**setLastAccessTime**(FileTime time)	Sets the last access time of the entry.
ZipEntry	**setLastModifiedTime**(FileTime time)	Sets the last modification time of the entry.
void	**setMethod**(int method)	Sets the compression method for the entry.
void	**setSize**(long size)	Sets the uncompressed size of the entry data.
void	**setTime**(long time)	Sets the last modification time of the entry.
void	**setTimeLocal**(LocalDateTime time)	Sets the last modification time of the entry in local date-time.
String	**toString**()	Returns a string representation of the ZIP entry.

Methods declared in class Object

equals, finalize, getClass, notify, notifyAll, wait, wait, wait

Field Details

STORED

```
public static final int STORED
```

Compression method for uncompressed entries.

See Also:

Constant Field Values

DEFLATED

```
public static final int DEFLATED
```

Compression method for compressed (deflated) entries.

See Also:

Constant Field Values

LOCSIG

```
static final long LOCSIG
```

Local file (LOC) header signature.

See Also:

Constant Field Values

EXTSIG

```
static final long EXTSIG
```

Extra local (EXT) header signature.

See Also:

Constant Field Values

CENSIG

```
static final long CENSIG
```

Central directory (CEN) header signature.

See Also:

Constant Field Values

ENDSIG

```
static final long ENDSIG
```

End of central directory (END) header signature.

See Also:

Constant Field Values

LOCHDR

```
static final int LOCHDR
```

Local file (LOC) header size in bytes (including signature).

See Also:

Constant Field Values

EXTHDR

`static final int EXTHDR`

Extra local (EXT) header size in bytes (including signature).

See Also:

Constant Field Values

CENHDR

`static final int CENHDR`

Central directory (CEN) header size in bytes (including signature).

See Also:

Constant Field Values

ENDHDR

`static final int ENDHDR`

End of central directory (END) header size in bytes (including signature).

See Also:

Constant Field Values

LOCVER

`static final int LOCVER`

Local file (LOC) header version needed to extract field offset.

See Also:

Constant Field Values

LOCFLG

`static final int LOCFLG`

Local file (LOC) header general purpose bit flag field offset.

See Also:

Constant Field Values

LOCHOW

```
static final int LOCHOW
```

Local file (LOC) header compression method field offset.

See Also:

Constant Field Values

LOCTIM

```
static final int LOCTIM
```

Local file (LOC) header modification time field offset.

See Also:

Constant Field Values

LOCCRC

```
static final int LOCCRC
```

Local file (LOC) header uncompressed file crc-32 value field offset.

See Also:

Constant Field Values

LOCSIZ

```
static final int LOCSIZ
```

Local file (LOC) header compressed size field offset.

See Also:

Constant Field Values

LOCLEN

```
static final int LOCLEN
```

Local file (LOC) header uncompressed size field offset.

See Also:

Constant Field Values

LOCNAM

```
static final int LOCNAM
```

Local file (LOC) header filename length field offset.

See Also:

Constant Field Values

LOCEXT

`static final int LOCEXT`

Local file (LOC) header extra field length field offset.

See Also:

Constant Field Values

EXTCRC

`static final int EXTCRC`

Extra local (EXT) header uncompressed file crc-32 value field offset.

See Also:

Constant Field Values

EXTSIZ

`static final int EXTSIZ`

Extra local (EXT) header compressed size field offset.

See Also:

Constant Field Values

EXTLEN

`static final int EXTLEN`

Extra local (EXT) header uncompressed size field offset.

See Also:

Constant Field Values

CENVEM

`static final int CENVEM`

Central directory (CEN) header version made by field offset.

See Also:

Constant Field Values

CENVER

`static final int CENVER`

Central directory (CEN) header version needed to extract field offset.

See Also:

Constant Field Values

CENFLG

`static final int CENFLG`

Central directory (CEN) header encrypt, decrypt flags field offset.

See Also:

Constant Field Values

CENHOW

`static final int CENHOW`

Central directory (CEN) header compression method field offset.

See Also:

Constant Field Values

CENTIM

`static final int CENTIM`

Central directory (CEN) header modification time field offset.

See Also:

Constant Field Values

CENCRC

`static final int CENCRC`

Central directory (CEN) header uncompressed file crc-32 value field offset.

See Also:

Constant Field Values

CENSIZ

`static final int CENSIZ`

Central directory (CEN) header compressed size field offset.

See Also:

Constant Field Values

CENLEN

`static final int CENLEN`

Central directory (CEN) header uncompressed size field offset.

See Also:

Constant Field Values

CENNAM

`static final int CENNAM`

Central directory (CEN) header filename length field offset.

See Also:

Constant Field Values

CENEXT

`static final int CENEXT`

Central directory (CEN) header extra field length field offset.

See Also:

Constant Field Values

CENCOM

`static final int CENCOM`

Central directory (CEN) header comment length field offset.

See Also:

Constant Field Values

CENDSK

`static final int CENDSK`

Central directory (CEN) header disk number start field offset.

See Also:

Constant Field Values

CENATT

```
static final int CENATT
```

Central directory (CEN) header internal file attributes field offset.

See Also:

Constant Field Values

CENATX

```
static final int CENATX
```

Central directory (CEN) header external file attributes field offset.

See Also:

Constant Field Values

CENOFF

```
static final int CENOFF
```

Central directory (CEN) header LOC header offset field offset.

See Also:

Constant Field Values

ENDSUB

```
static final int ENDSUB
```

End of central directory (END) header number of entries on this disk field offset.

See Also:

Constant Field Values

ENDTOT

```
static final int ENDTOT
```

End of central directory (END) header total number of entries field offset.

See Also:

Constant Field Values

ENDSIZ

```
static final int ENDSIZ
```

End of central directory (END) header central directory size in bytes field offset.

See Also:

Constant Field Values

ENDOFF

`static final int ENDOFF`

End of central directory (END) header offset for the first CEN header field offset.

See Also:

Constant Field Values

ENDCOM

`static final int ENDCOM`

End of central directory (END) header ZIP file comment length field offset.

See Also:

Constant Field Values

Constructor Details

ZipEntry

`public ZipEntry(String name)`

Creates a new ZIP entry with the specified name.

Parameters:

`name` - The entry name

Throws:

`NullPointerException` - if the entry name is null

`IllegalArgumentException` - if the combined length of the entry name and the CEN Header size exceeds 65,535 bytes.

ZipEntry

`public ZipEntry(ZipEntry e)`

Creates a new ZIP entry with fields taken from the specified ZIP entry.

Parameters:

`e` - A ZIP Entry object

Throws:

`NullPointerException` - if the entry object is null

Method Details

getName

`public String getName()`

Returns the name of the entry.

Returns:

the name of the entry

setTime

`public void setTime(long time)`

Sets the last modification time of the entry.

If the entry is output to a ZIP file or ZIP file formatted output stream the last modification time set by this method will be stored into the `date and time fields` of the ZIP file entry and encoded in standard `MS-DOS date and time format`. The `default TimeZone` is used to convert the epoch time to the MS-DOS date and time.

Parameters:

`time` - The last modification time of the entry in milliseconds since the epoch

See Also:

`getTime()`, `getLastModifiedTime()`

getTime

`public long getTime()`

Returns the last modification time of the entry.

If the entry is read from a ZIP file or ZIP file formatted input stream, this is the last modification time from the `date and time fields` of the ZIP file entry. The `default TimeZone` is used to convert the standard MS-DOS formatted date and time to the epoch time.

Returns:

The last modification time of the entry in milliseconds since the epoch, or -1 if not specified

See Also:

`setTime(long)`, `setLastModifiedTime(FileTime)`

setTimeLocal

```
public void setTimeLocal(LocalDateTime time)
```

Sets the last modification time of the entry in local date-time.

If the entry is output to a ZIP file or ZIP file formatted output stream the last modification time set by this method will be stored into the `date and time fields` of the ZIP file entry and encoded in standard `MS-DOS date and time format`. If the date-time set is out of the range of the standard `MS-DOS date and time format`, the time will also be stored into ZIP file entry's extended timestamp fields in `optional extra data` in UTC time. The `system default TimeZone` is used to convert the local date-time to UTC time.

`LocalDateTime` uses a precision of nanoseconds, whereas this class uses a precision of milliseconds. The conversion will truncate any excess precision information as though the amount in nanoseconds was subject to integer division by one million.

Parameters:

`time` - The last modification time of the entry in local date-time

Throws:

`NullPointerException` - if `time` is null

Since:

9

See Also:

`getTimeLocal()`

getTimeLocal

```
public LocalDateTime getTimeLocal()
```

Returns the last modification time of the entry in local date-time.

If the entry is read from a ZIP file or ZIP file formatted input stream, this is the last modification time from the zip file entry's `optional extra data` if the extended timestamp fields are present. Otherwise, the last modification time is read from entry's standard MS-DOS formatted `date and time fields`.

The `system default TimeZone` is used to convert the UTC time to local date-time.

Returns:

The last modification time of the entry in local date-time

Since:

9

See Also:

`setTimeLocal(LocalDateTime)`

setLastModifiedTime

```
public ZipEntry setLastModifiedTime(FileTime time)
```

Sets the last modification time of the entry.

When output to a ZIP file or ZIP file formatted output stream the last modification time set by this method will be stored into ZIP file entry's `date and time fields` in `standard MS-DOS date and time format`), and the extended timestamp fields in `optional extra data` in UTC time.

Parameters:

`time` - The last modification time of the entry

Returns:

This ZIP entry

Throws:

`NullPointerException` - if the `time` is null

Since:

1.8

See Also:

`getLastModifiedTime()`

getLastModifiedTime

```
public FileTime getLastModifiedTime()
```

Returns the last modification time of the entry.

If the entry is read from a ZIP file or ZIP file formatted input stream, this is the last modification time from the zip file entry's `optional extra data` if the extended timestamp fields are present. Otherwise the last modification time is read from the entry's `date and time fields`, the `default TimeZone` is used to convert the standard MS-DOS formatted date and time to the epoch time.

Returns:

The last modification time of the entry, null if not specified

Since:

1.8

See Also:

setLastModifiedTime(FileTime)

setLastAccessTime

`public ZipEntry setLastAccessTime(FileTime time)`

Sets the last access time of the entry.

If set, the last access time will be stored into the extended timestamp fields of entry's `optional extra data`, when output to a ZIP file or ZIP file formatted stream.

Parameters:

`time` - The last access time of the entry

Returns:

This ZIP entry

Throws:

`NullPointerException` - if the `time` is null

Since:

1.8

See Also:

getLastAccessTime()

getLastAccessTime

`public FileTime getLastAccessTime()`

Returns the last access time of the entry.

The last access time is from the extended timestamp fields of entry's `optional extra data` when read from a ZIP file or ZIP file formatted stream.

Returns:

The last access time of the entry, null if not specified

Since:

1.8

See Also:

setLastAccessTime(FileTime)

setCreationTime

`public ZipEntry setCreationTime(FileTime time)`

Sets the creation time of the entry.

If set, the creation time will be stored into the extended timestamp fields of entry's `optional extra data`, when output to a ZIP file or ZIP file formatted stream.

Parameters:

`time` - The creation time of the entry

Returns:

This ZIP entry

Throws:

`NullPointerException` - if the `time` is null

Since:

1.8

See Also:

`getCreationTime()`

getCreationTime

```
public FileTime getCreationTime()
```

Returns the creation time of the entry.

The creation time is from the extended timestamp fields of entry's `optional extra data` when read from a ZIP file or ZIP file formatted stream.

Returns:

the creation time of the entry, null if not specified

Since:

1.8

See Also:

`setCreationTime(FileTime)`

setSize

```
public void setSize(long size)
```

Sets the uncompressed size of the entry data.

Parameters:

`size` - the uncompressed size in bytes

Throws:

`IllegalArgumentException` - if the specified size is less than 0, is greater than 0xFFFFFFFF when ZIP64 format is not supported, or is less than 0 when ZIP64 is supported

See Also:

`getSize()`

getSize

```
public long getSize()
```

Returns the uncompressed size of the entry data.

Returns:

the uncompressed size of the entry data, or -1 if not known

See Also:

`setSize(long)`

getCompressedSize

```
public long getCompressedSize()
```

Returns the size of the compressed entry data.

In the case of a stored entry, the compressed size will be the same as the uncompressed size of the entry.

Returns:

the size of the compressed entry data, or -1 if not known

See Also:

`setCompressedSize(long)`

setCompressedSize

```
public void setCompressedSize(long csize)
```

Sets the size of the compressed entry data.

Parameters:

`csize` - the compressed size to set

See Also:

`getCompressedSize()`

setCrc

```
public void setCrc(long crc)
```

Sets the CRC-32 checksum of the uncompressed entry data.

Parameters:

`crc` - the CRC-32 value

Throws:

`IllegalArgumentException` - if the specified CRC-32 value is less than 0 or greater than 0xFFFFFFFF

See Also:

`getCrc()`

getCrc

```
public long getCrc()
```

Returns the CRC-32 checksum of the uncompressed entry data.

Returns:

the CRC-32 checksum of the uncompressed entry data, or -1 if not known

See Also:

`setCrc(long)`

setMethod

```
public void setMethod(int method)
```

Sets the compression method for the entry.

Parameters:

`method` - the compression method, either STORED or DEFLATED

Throws:

`IllegalArgumentException` - if the specified compression method is invalid

See Also:

`getMethod()`

getMethod

```
public int getMethod()
```

Returns the compression method of the entry.

Returns:

the compression method of the entry, or -1 if not specified

See Also:

`setMethod(int)`

setExtra

```
public void setExtra(byte[] extra)
```

Sets the optional extra field data for the entry.

Invoking this method may change this entry's last modification time, last access time and creation time, if the `extra` field data includes the extensible timestamp fields, such as `NTFS tag 0x0001` or `Info-ZIP Extended Timestamp`, as specified in Info-ZIP Application Note 970311.

Parameters:

`extra` - The extra field data bytes

Throws:

`IllegalArgumentException` - if the combined length of the specified extra field data, the entry name, the entry comment, and the CEN Header size exceeds 65,535 bytes.

See Also:

`getExtra()`

getExtra

```
public byte[] getExtra()
```

Returns the extra field data for the entry.

Returns:

the extra field data for the entry, or null if none

See Also:

`setExtra(byte[])`

setComment

```
public void setComment(String comment)
```

Sets the optional comment string for the entry. If `comment` is an empty string or `null` then the entry will have no comment.

Parameters:

`comment` - the comment string, or an empty string or null for no comment

Throws:

`IllegalArgumentException` - if the combined length of the specified entry comment, the entry name, the extra field data, and the CEN Header size exceeds 65,535 bytes.

See Also:

`getComment()`

getComment

```
public String getComment()
```

Returns the comment string for the entry.

Returns:

the comment string for the entry, or null if none

See Also:

`setComment(String)`

isDirectory

```
public boolean isDirectory()
```

Returns true if this is a directory entry. A directory entry is defined to be one whose name ends with a '/'.

Returns:

true if this is a directory entry

toString

```
public String toString()
```

Returns a string representation of the ZIP entry.

Overrides:

`toString` in class `Object`

Returns:

a string representation of the object

hashCode

```
public int hashCode()
```

Returns the hash code value for this entry.

Overrides:

`hashCode` in class `Object`

Returns:

a hash code value for this object

See Also:

`Object.equals(java.lang.Object)`,

System.identityHashCode(Object)

clone

```
public Object clone()
```

Returns a copy of this entry.

Overrides:

clone in class Object

Returns:

a clone of this instance.

See Also:

Cloneable

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
Java is a trademark or registered trademark of Oracle and/or its affiliates in the US and other countries.
Copyright © 1993, 2026, Oracle and/or its affiliates, 500 Oracle Parkway, Redwood Shores, CA 94065 USA.
All rights reserved. Use is subject to license terms and the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.